Shiyu Chen

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Robotics perception — LiDAR–camera fusion & calibration · 3D semantic mapping · C++/ROS 2 · deep learning on edge GPUs

EDUCATION

Carnegie Mellon University — M.S. in Mechanical Engineering (Research) Aug 2025 – May 2027 (expected)
Computational Engineering & Robotics Lab (CERLAB), advisor Prof. Kenji Shimada Pittsburgh, PA

University of Cincinnati & Chongqing University — B.S. / B.Eng. in Mechanical Engineering (joint program) 2020 – 2025
Robotics Minor · GPA 3.71/4.0 (UC) · Dean's List ×4 Cincinnati, OH / Chongqing, CN

SKILLS

Languages	C++, Python, MATLAB, C
Robotics	ROS 1/2, Isaac Sim, Spot SDK, SLAM, LiDAR–camera calibration, stereo vision, viewpoint & path planning
Perception / ML	PyTorch, TensorRT, OpenCV, YOLO, SAM2, VLM APIs, point-cloud processing (ICP)
Hardware / Tools	Jetson Xavier NX / AGX Orin, Huawei Ascend, Ouster LiDAR, Linux, Docker, Git, SolidWorks, 3D printing

EXPERIENCE

Graduate Research Assistant, CERLAB, Carnegie Mellon University Aug 2025 – Present · Pittsburgh, PA

Semantic exploration for a legged inspection robot (CMU × YKK factory-automation project · co-first author, ICRA 2027 submission)

- Developed **5-DoF viewpoint planning** for Spot: body pitch/roll on top of planar motion reveals surfaces a level camera misses.
- Built 3D semantic mapping: per-instance detection masks back-projected into 3D object boxes in the occupancy map.
- Sampled body postures by expected coverage gain; aim-aligned execution avoids needless reorientation.
- Added VLM-assisted viewpoint pruning (observation history + bird's-eye-view map), with a geometric fallback.
- Built the Isaac Sim benchmark: three industrial scenes, ground-truth surfaces, plug-in planners, Dockerized; planner in C++/ROS.
- **+8–10 pp coverage**, **17–32% less exploration time** vs. the 3-DoF baseline; **53–73% fewer postures** than other 5-DoF methods.
- Deployed on a real Spot in a machine shop (five target machines).

Real-time LiDAR–camera fusion on Spot

- Fused Ouster LiDAR point clouds with Insta360 X4 panoramic images for semantic registration.
- **10 FPS** at full LiDAR rate, **<100 ms** latency, **<10 px** reprojection error.
- Designed a 3D-printed rigid sensor mount; registration stays stable under gait vibration.

Shoe-interior 3D scanner (Sep 2026 – Present · sole researcher)

- Geometry-only 3D reconstruction of the inside of a shoe for fit checks (e.g., diabetic feet); scope extended to garments.
- Designed a stereo endoscopic probe: two global-shutter cameras, **34 mm baseline, 148° FOV**, LED ring, wall fiducials.
- Error budget: ≈0.3 mm depth error at 100 mm standoff; **target reconstruction error ≤ 1 mm**.
- Designed a rail fixture for point-by-point scans; test plan on 3D-printed cavity halves with CAD ground truth.

Computer Vision R&D Assistant, CISDI Information Technology May 2022 – Dec 2023 · Chongqing, China

- Trained PyTorch object detectors for belt-conveyor inspection.
- Deployed them on NVIDIA Jetson Xavier NX / AGX Orin with **TensorRT**, cutting inference latency by **40%**.
- Ported the Python inference code to C++.
- Adapted the models to Huawei Ascend NPUs; **2nd prize**, Ascend AI competition.
- Built a PyQt annotation tool for **100k+** images. Received a return offer.
- Co-inventor on **4 granted Chinese invention patents** (2025–2026) in deep-learning vision for steelmaking: steel-stream landing point, slag splashing, slag discharge.
- 2 further patent applications pending in belt-conveyor inspection.

Undergraduate Researcher, University of Cincinnati (Prof. Janet Dong) Sep 2024 – May 2025 · Cincinnati, OH

Cable-driven soft robotic arm for minimally invasive surgery

- Designed and prototyped a wire-actuated continuum arm: **0.27 mm/step** resolution, **>230°** bending range.
- Modeled nonlinear actuation; led hardware–control integration.

PUBLICATIONS

X. Zhan*, **S. Chen***, K. Shimada, "Pose-aware Legged Robot Semantic Exploration with Omnidirectional Perception in Confined Unknown Environments," *arXiv:2609.19460*, 2026. Under review at ICRA 2027. *Equal contribution.

Y. Wang, C. Liu, K. Jiang, B. Wu, **S. Chen**, J. Dong, A. Ashfaq, "Innovative Flexible Robotic Arm for Enhanced Precision in Transaortic Surgical Myectomy," *ASME J. Medical Diagnostics*, 9(2): 021001, 2026.

L. Liu, **S. Chen**, Z. Tang, "Intelligent Diagnosis of Bearing Faults Based on SDP Image Fusion," *ICICML*, Chengdu, 2023.